STACKS, EXAMPLES OF DEFORMATIONS: LEMMA-FINITE-TYPE-RINGS-TI

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1123-1132 of sources/stacks-examples-defos/pinned-original-source.tex; proof span is lines 1134-1195, the complete author proof (60 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 3 of queue_examples-defos.json (pb=2941, stmt 1123-1132, proof 1134-1195).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 00TT: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00RY: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00RT: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0ALW: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0583: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00S7: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00J3: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 07BU: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02CE: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 00FV: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DY3: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1072–1079. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 087R: sources/stacks-examples-defos/pinned-original-source.tex, source lines 1195–1195. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *In Lemma 0DY3 if P is a finite type k-algebra, then*

(1) *$Inf(\mathcal{D}ef_P)$ is finite dimensional if and only if $\dim(P) = 0$, and*
(2) *$T\mathcal{D}ef_P$ is finite dimensional if $\mathrm{Spec}(P) \to \mathrm{Spec}(k)$ is smooth except at a finite number of points.*

Proof. Proof of (1). We view $\mathrm{Der}_k(P, P)$ as a P-module. If it has finite dimension over k, then it has finite length as a P-module, hence it is supported in finitely many closed points of $\mathrm{Spec}(P)$ (Algebra, Lemma 00J3). Since $\mathrm{Der}_k(P, P) = \mathrm{Hom}_P(\Omega_{P/k}, P)$ we see that $\mathrm{Der}_k(P, P)_\mathfrak{p} = \mathrm{Der}_k(P_\mathfrak{p}, P_\mathfrak{p})$ for any prime $\mathfrak{p} \subset P$ (this uses Algebra, Lemmas 00RT, 00RY, and 0583). Let $\mathfrak{p}$ be a minimal prime ideal of P corresponding to an irreducible component of dimension $d > 0$. Then $P_\mathfrak{p}$ is an Artinian local ring essentially of finite type over k with residue field and $\Omega_{P_\mathfrak{p}/k}$ is nonzero for example by Algebra, Lemma 00TT. Any nonzero finite module over an Artinian local ring has both a sub and a quotient module isomorphic to the residue field. Thus we find that $\mathrm{Der}_k(P_\mathfrak{p}, P_\mathfrak{p}) = \mathrm{Hom}_{P_\mathfrak{p}}(\Omega_{P_\mathfrak{p}/k}, P_\mathfrak{p})$ is nonzero too. Combining all of the above we find that (1) is true.

Proof of (2). For a prime $\mathfrak{p}$ of P we will use that $NL_{P_\mathfrak{p}/k} = (NL_{P/k})_\mathfrak{p}$ (Algebra, Lemma 00S7) and we will use that $\mathrm{Ext}^1_P(NL_{P/k}, P)_\mathfrak{p} = \mathrm{Ext}^1_{P_\mathfrak{p}}(NL_{P_\mathfrak{p}/k}, P_\mathfrak{p})$ (More on Algebra, Lemma 087R). Given a prime $\mathfrak{p} \subset P$ then $k \to P$ is smooth at $\mathfrak{p}$ if and only if $(NL_{P/k})_\mathfrak{p}$ is quasi-isomorphic to a finite projective module placed in degree 0 (this follows immediately from the definition of a smooth ring map but it also follows from the stronger Algebra, Lemma 07BU).

Assume that P is smooth over k at all but finitely many primes. Then these "bad" primes are maximal ideals $\mathfrak{m}_1, \ldots, \mathfrak{m}_n \subset P$ by Algebra, Lemma 0ALW and the fact that the "bad" primes form a closed subset of $\mathrm{Spec}(P)$. For $\mathfrak{p} \notin \{\mathfrak{m}_1, \ldots, \mathfrak{m}_n\}$ we have $\mathrm{Ext}^1_P(NL_{P/k}, P)_\mathfrak{p} = 0$ by the results above. Thus $\mathrm{Ext}^1_P(NL_{P/k}, P)$ is a finite P-module whose support is contained in $\{\mathfrak{m}_1, \ldots, \mathfrak{m}_r\}$. By Algebra, Proposition 02CE for example, we find that the dimension over k of $\mathrm{Ext}^1_P(NL_{P/k}, P)$ is a finite integer combination of $\dim_k \kappa(\mathfrak{m}_i)$ and hence finite by the Hilbert Nullstellensatz (Algebra, Theorem 00FV). $\square$